CIE AS PHYSICS (9702) • QUESTION PAPER

10.2 Kirchhoff's Laws

Syllabus Section: 10. D.C. Circuits **Paper Type:** Multiple Choice (Paper 1) **Total:** 18 Questions (18 Marks)

Question 1 Easy [1 mark]

The principles of conservation of which two quantities are associated with Kirchhoff's first and second Laws?

	First Law	Second Law
A	charge	energy
B	charge	voltage
C	energy	charge
D	voltage	charge

A

B

C

D

Question 2 Easy [1 mark]

In deriving a formula for the combined resistance of three different resistors in series, Kirchhoff's Laws are used.

Which physics principle is involved in this derivation ?

A the conservation of charge

B the direction of the flow of charge is from negative to positive

C the potential difference across each resistor is the same

D the current varies in each resistor, in proportion to the resistor value

Question 3 Easy [1 mark]

The combined resistance R_T of two resistors of resistances R_1 and R_2 connected in parallel is given by the formula

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$$

Which statement is used in the derivation of this formula?

A the currents through the two resistors are equal

B the potential difference across each resistor is the same

C the supply current is split between the two resistors in the same ratio as the ratio of their resistances

D the total power dissipated is the sum of the powers dissipated in the two resistors separately

Question 4 Easy [1 mark]

Each of Kirchhoff's two Laws presumes that some quantity is conserved.

Which row states Kirchhoff's **first** Law and names the quantity that is conserved ?

	Statement	Quantity
A	The algebraic sum of currents into a junction is zero	charge
B	The algebraic sum of currents into a junction is zero	energy
C	The e. m. f. in a loop is equal to the algebraic sum of the product of current and resistance around the loop	charge
D	The e .m .f. in a loop is equal to the algebraic sum of the product of current and resistance around the loop	energy

A

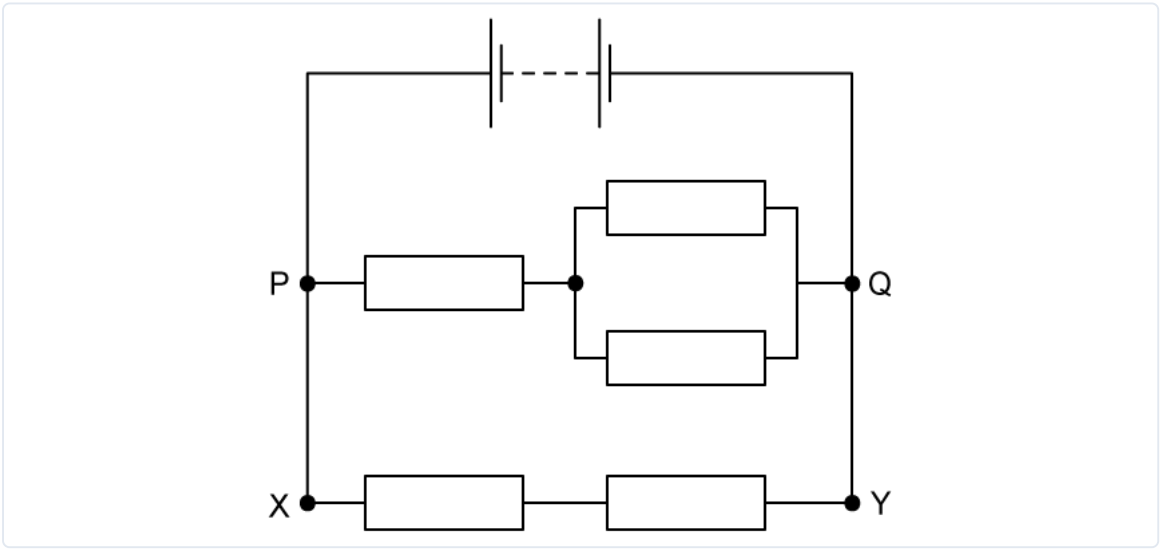
B

C

D

Question 5 Easy [1 mark]

A circuit consisting of five **identical** resistors is shown:



Which of the following statements is **false**?

A The potential difference between **P** and **Q** is equal to the potential difference between **X** and **Y**

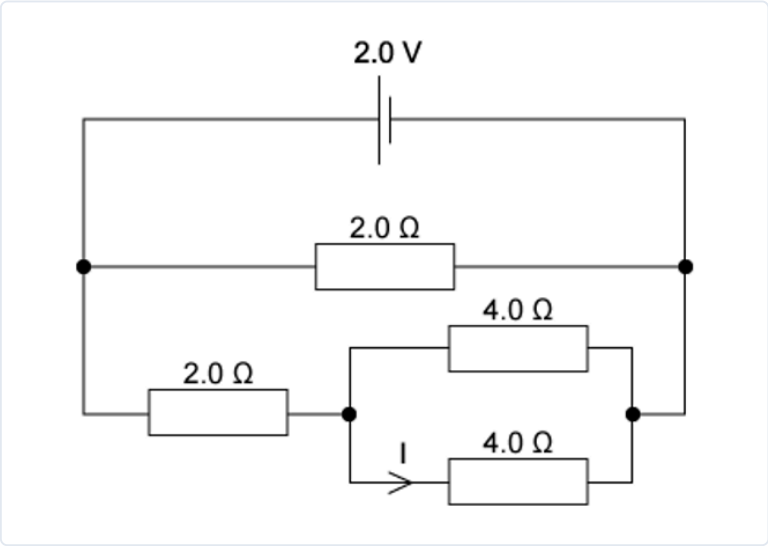
B The total resistance between **X** and **Y** is equal to the sum of the resistances between **X** and **Y**

C The total resistance between **P** and **Q** is equal to the sum of the resistances along **P** and **Q**

D There is less total resistance between **P** and **Q** than there is between **X** and **Y**

Question 6 Medium [1 mark]

A cell of e .m .f. 2.0 V and negligible internal resistance is connected to a network of resistors as shown.



What is the current *I* ?

A 0.25 A

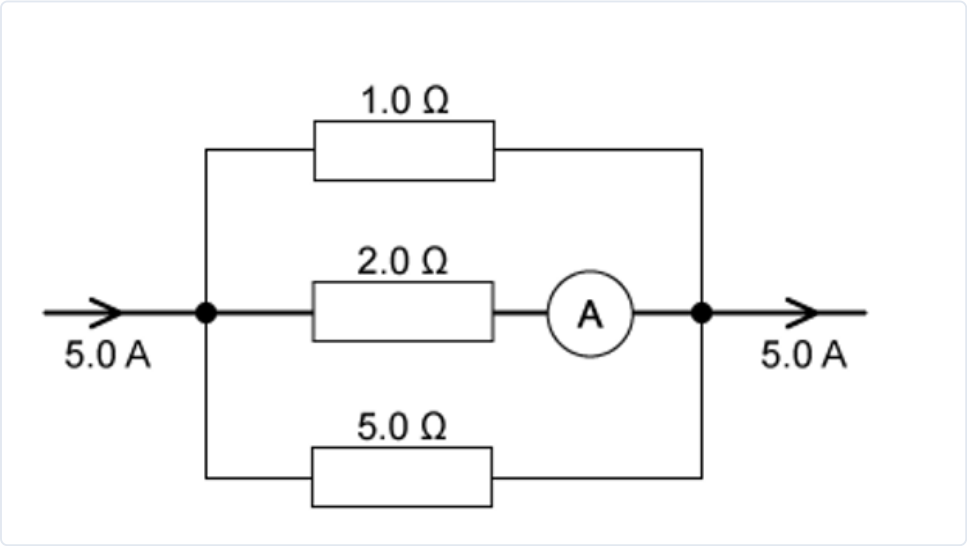
B 0.33 A

C 0.50 A

D 1.5 A

Question 7 Medium [1 mark]

The diagram shows part of a current-carrying circuit. The ammeter has negligible resistance.



What is the reading on the ammeter ?

A 0.7 A

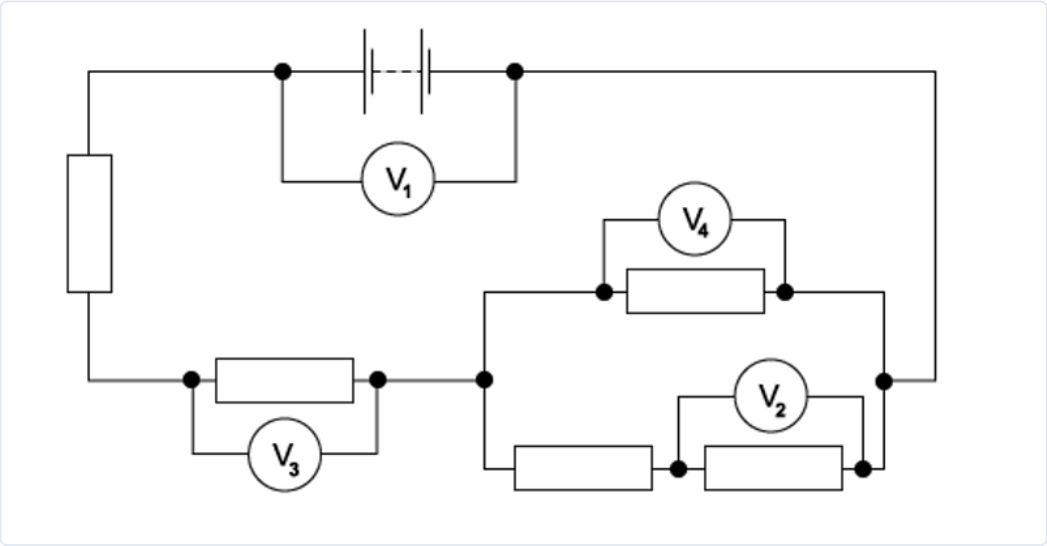
B 1.3 A

C 1.5 A

D 1.7 A

Question 8 Medium [1 mark]

In the circuit shown, all the resistors are identical.



The reading on voltmeter V_1 is 8.0 V and the reading on voltmeter V_2 is 1.0 V. What are the readings on the other voltmeters?

	Reading on voltmeter V_3 / V	Reading on voltmeter V_4 / V
A	1.5	1.0
B	3.0	2.0
C	4.5	3.0
D	6.0	4.0

A

B

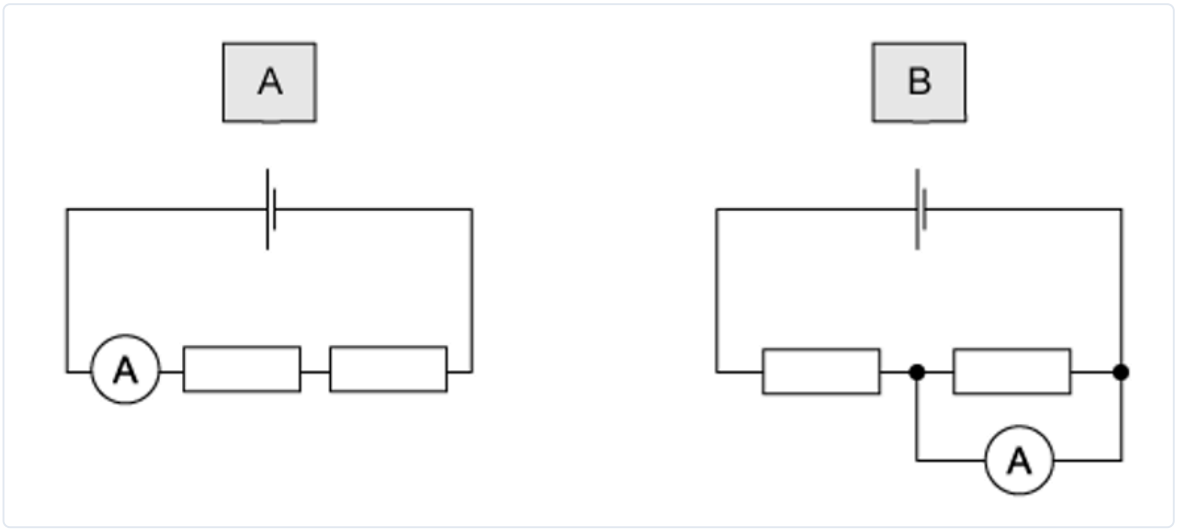
C

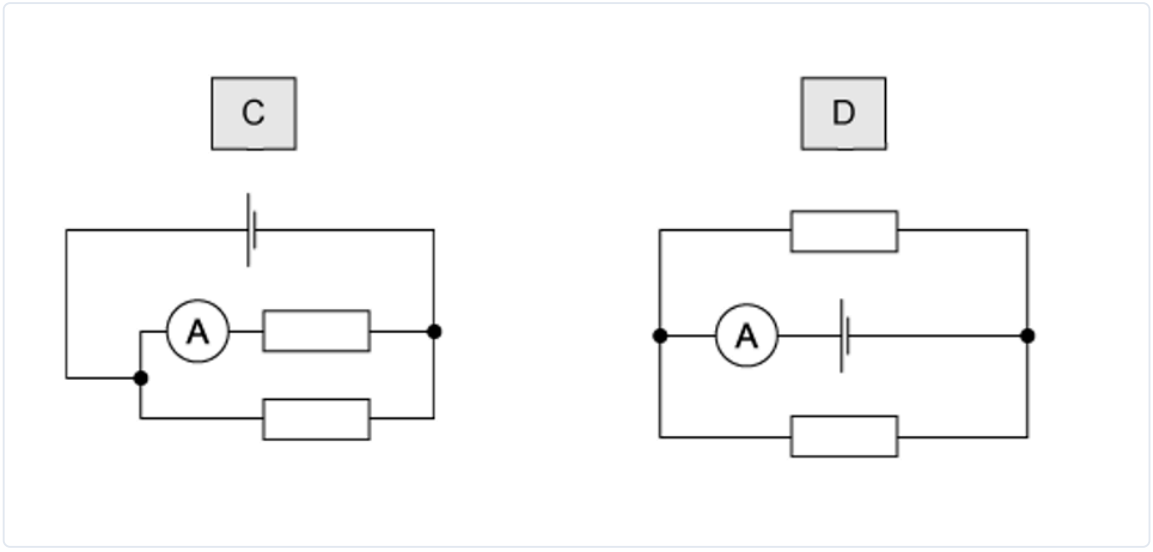
D

Question 9 Medium [1 mark]

A cell, two resistors of equal resistance and an ammeter are used to construct four circuits. The resistors are the only parts of the circuits that have resistance.

In which circuit will the ammeter show the greatest reading ?





A

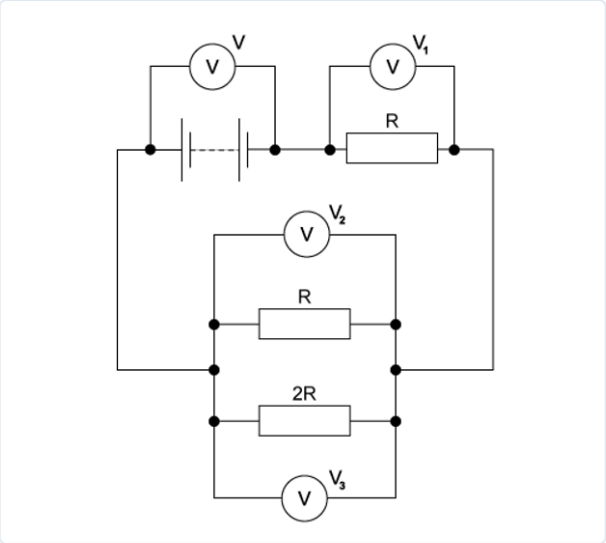
B

C

D

Question 10 Medium [1 mark]

The diagram shows a circuit with four voltmeter readings V, V_1, V_2 and V_3.



Which equation relating the voltmeter readings must be true?

A $V = V_1 + V_2 + V_3$

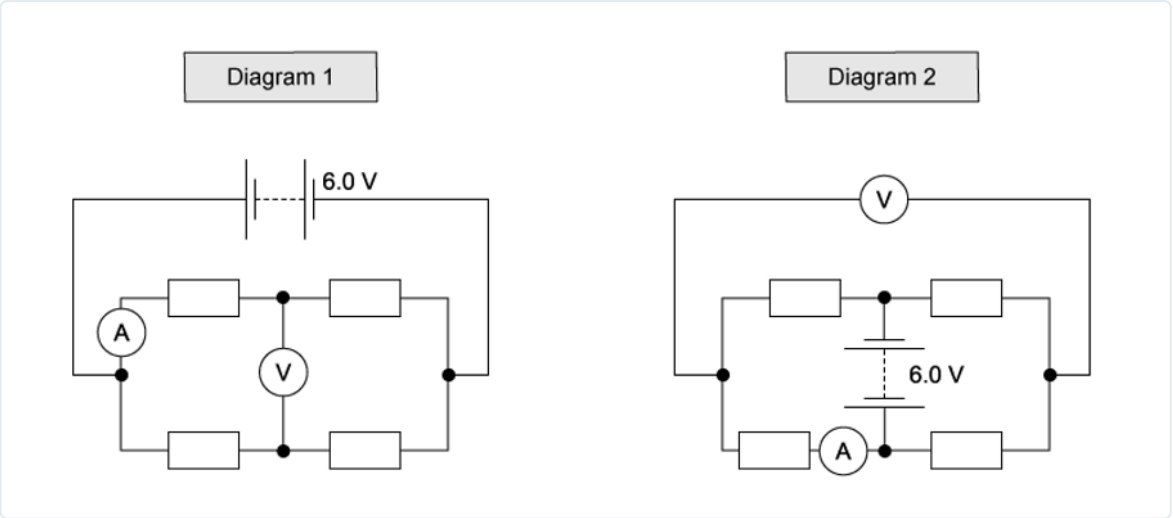
B $V + V_1 = V_2 + V_3$

C $V_3 = 2(V_2)$

D $V - V_1 = V_3$

Question 11 Medium [1 mark]

When four identical resistors are connected as shown in diagram 1, the ammeter reads 1.0 A, and the voltmeter reads zero.



The resistors and meters are reconnected to the supply, as shown in diagram 2.

What are the meter readings in diagram 2?

	Voltmeter reading / V	Ammeter reading / A
A	0	1.0
B	3.0	0.5
C	3.0	1.0
D	6.0	0

A

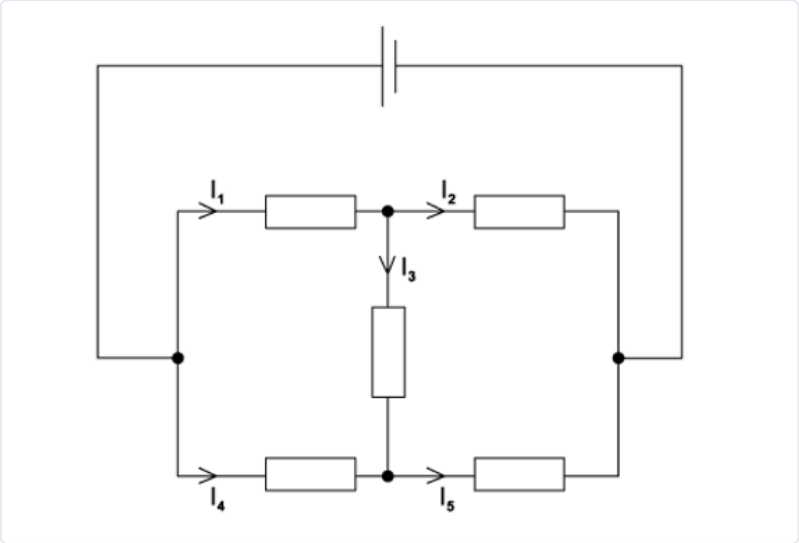
B

C

D

Question 12 Medium [1 mark]

The diagram shows currents I_1, I_2, I_3, I_4 and I_5 in different branches of a circuit.



Which one of the following is correct?

A $I_1 = I_2 + I_3$

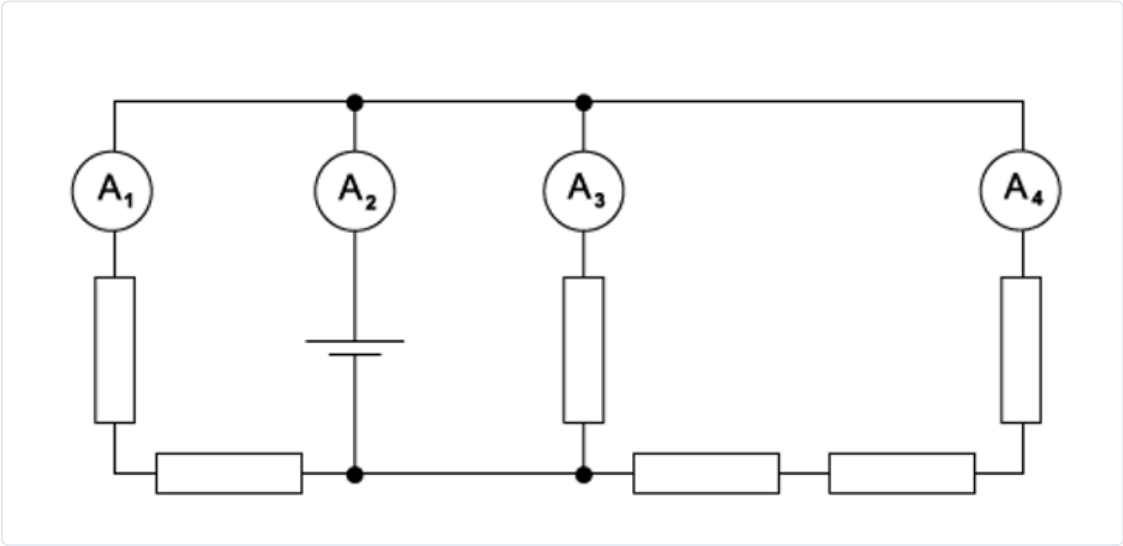
B $I_2 = I_1 + I_3$

C $I_3 = I_4 + I_5$

D $I_4 = I_5 + I_3$

Question 13 Medium [1 mark]

In the circuit shown, all the resistors are identical and all the ammeters have negligible resistance.



The reading on ammeter A_1 is 0.6 A.

What are the readings on the other ammeters?

	Reading on ammeter A_2 / A	Reading on ammeter A_3 / A	Reading on ammeter A_4 / A
A	1.0	0.3	0.1
B	1.4	0.6	0.2
C	1.8	0.9	0.3
D	2.2	1.2	0.4

A

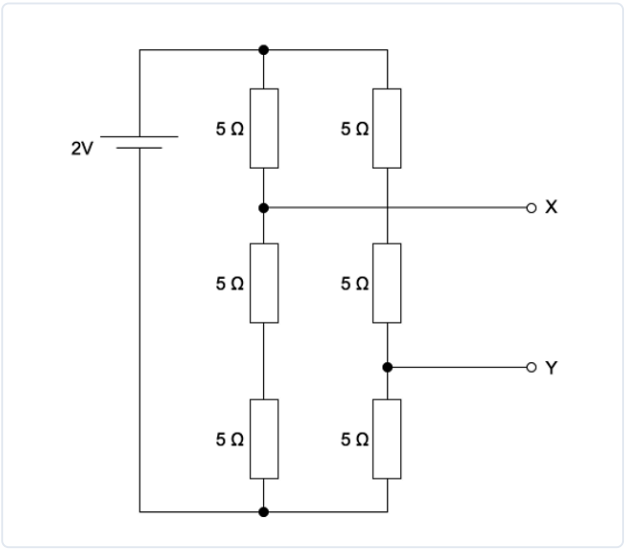
B

C

D

Question 14 Hard [1 mark]

Six resistors, each of resistance 5 Ω, are connected to a 2 V cell of negligible internal resistance.



What is the potential difference between terminals X and Y ?

A $\frac{2}{3}\ V$

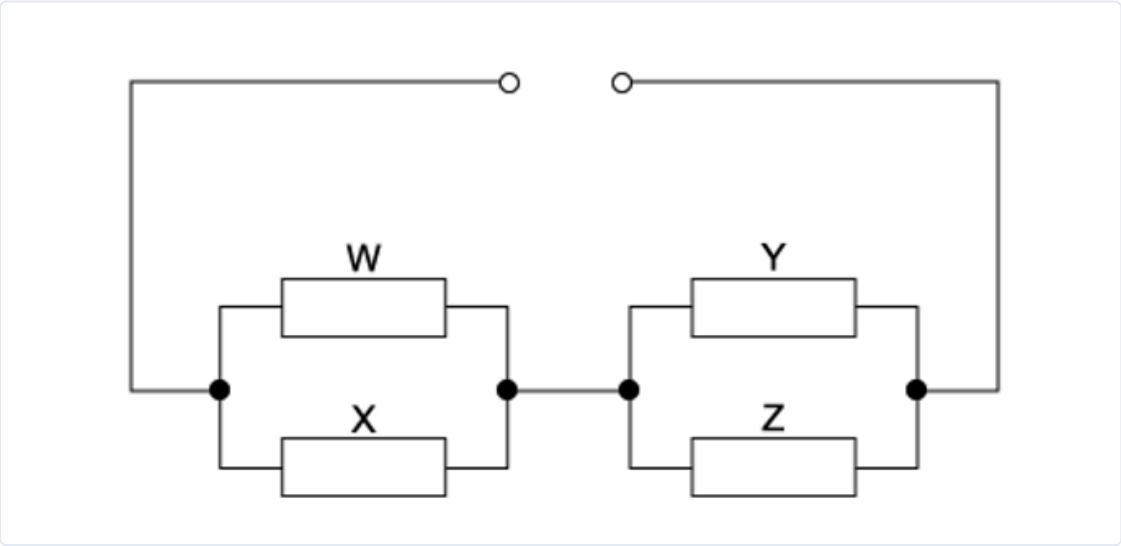
B $\frac{8}{9}\ V$

C $\frac{4}{3}\ V$

D $2\ V$

Question 15 Hard [1 mark]

Four resistors of equal value are connected as shown.



How will the powers to the resistors change when resistor W is removed?

A the powers to X, Y and Z will all increase

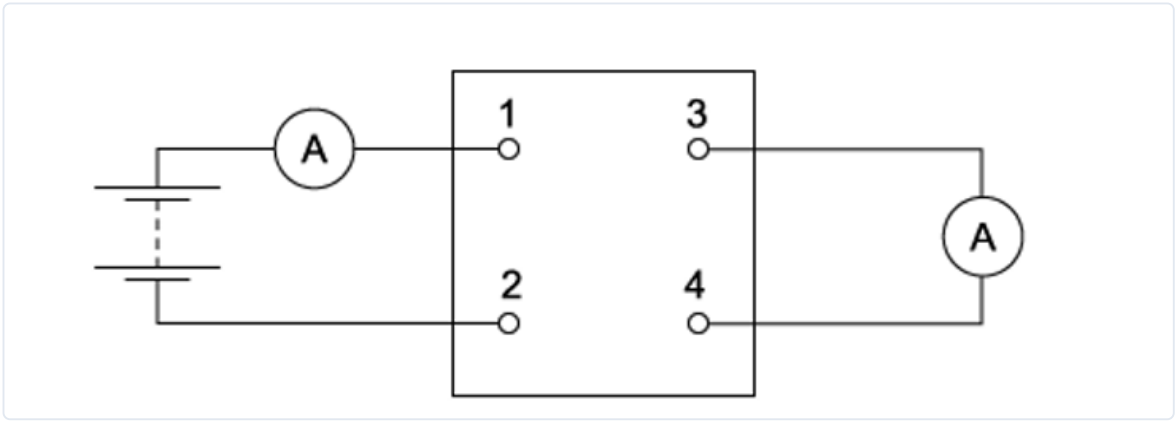
B the power to X will decrease and the powers to Y and Z will increase

C the power to X will increase and the powers to Y and Z will decrease

D the power to X will increase and the powers to Y and Z will remain unaltered

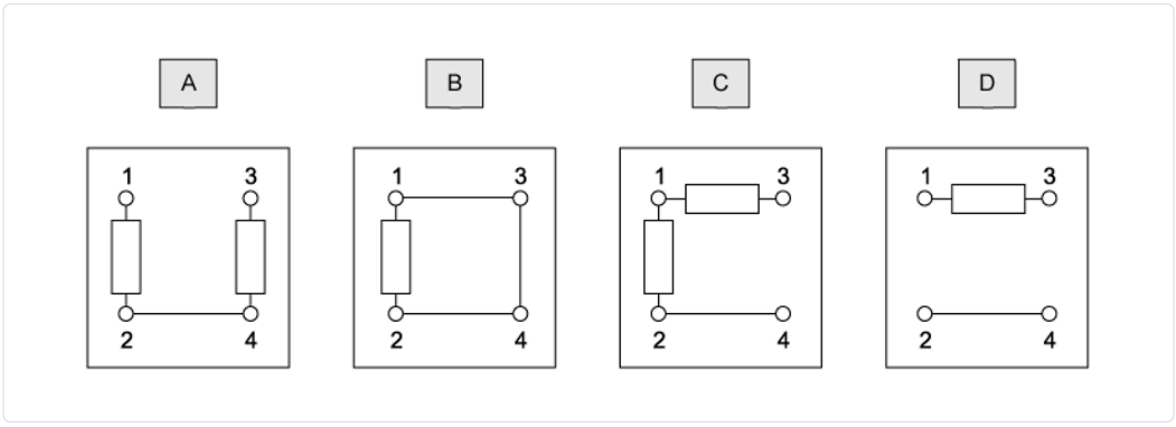
Question 16 Hard [1 mark]

The diagram shows a four-terminal box connected to a battery and two ammeters.



The currents in the two meters are identical.

Which circuit, within the box, will give this result ?



A

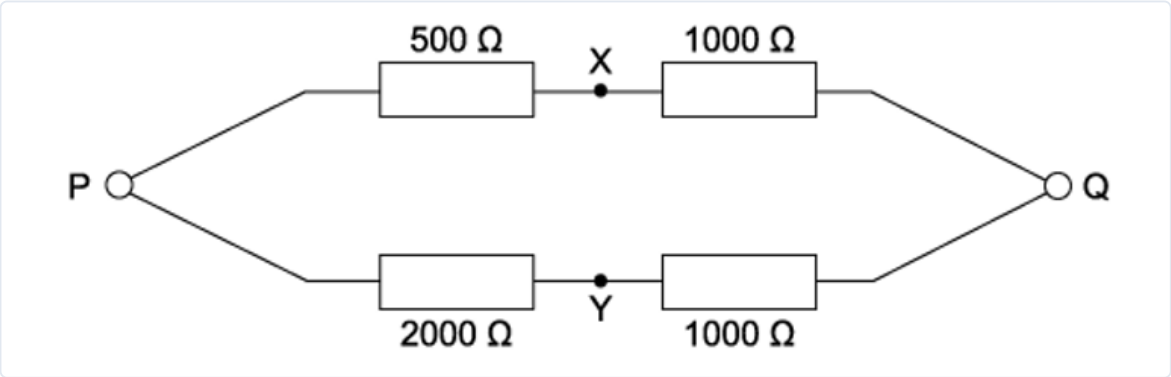
B

C

D

Question 17 Hard [1 mark]

A p. d. of 12 V is connected between P and Q.



What is the p. d. between X and Y?

A 0 V

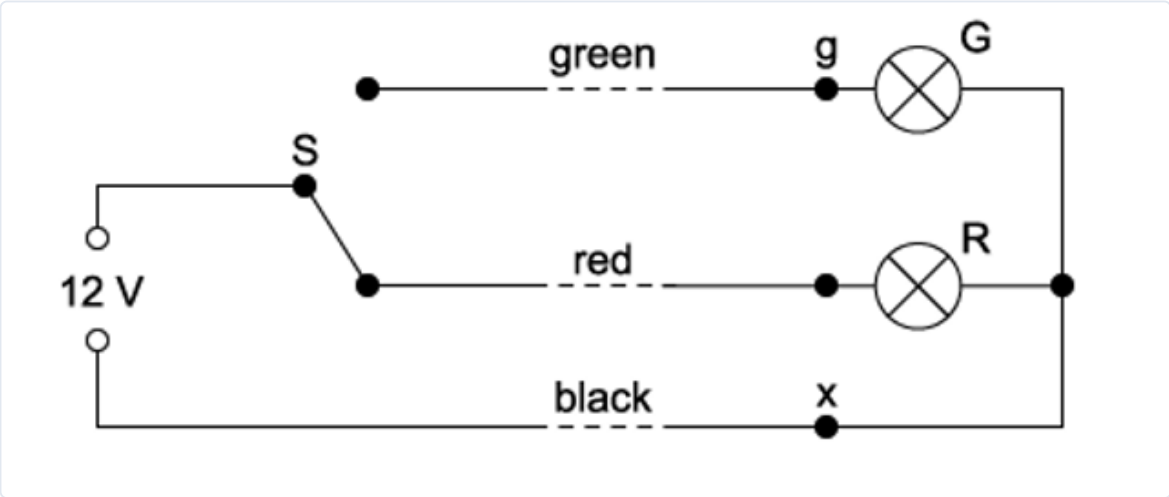
B 4 V

C 6 V

D 8 V

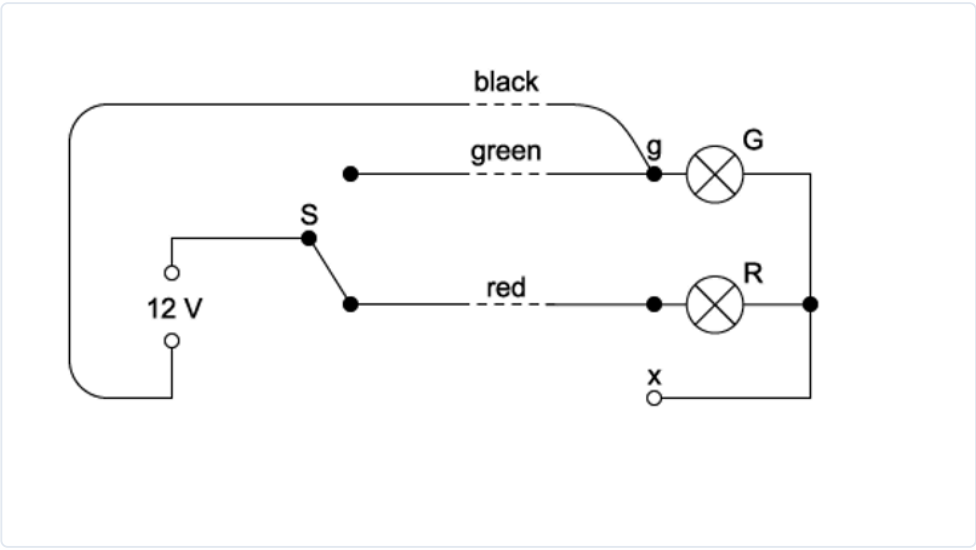
Question 18 Hard [1 mark]

The diagram shows the circuit for a signal to display a green or a red light. It is controlled by the switch S.



The signal is some way from S to which it is connected by a cable with green, red and black wires. At the signal, the green and red wires are connected to the corresponding lamp and the black wire is connected to a terminal x to provide a common return. The arrangement is shown correctly connected and with the switch set to illuminate the red lamp.

During maintenance, the wires at the signal are disconnected and, when reconnected, the black wire is connected in error to the green lamp (terminal g) instead of terminal x. The red wire is connected correctly to its lamp and connections at S remain as in the diagram.



When the system is tested with the switch connection to the red wire, what does the signal show ?

A the green lamp illuminated normally

B the red lamp illuminated normally

C the red and green lamps both illuminated normally

D the red and green lamps both illuminated dimly